

nD: 4, H: 1, H+: 1

term0

$$+ \left(\frac{1}{24} * a_t^2 - \frac{1}{4} * a_t * \hat{\mu} + \frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} - \frac{1}{2} * \frac{a_t * \hat{\mu}}{\varepsilon} \right) * M_s^{-2} \varepsilon_{i,j} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon^{\alpha_4, \alpha_3} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * \varepsilon_{\dot{\alpha}_4, \dot{\alpha}_3} * (D_{\dot{\alpha}_1}^{\dot{\alpha}_1} D_{\dot{\alpha}_2}^{\dot{\alpha}_2} H)_i * (D_{\dot{\alpha}_3}^{\dot{\alpha}_3} D_{\dot{\alpha}_4}^{\dot{\alpha}_4} H^\dagger)_j \quad (1)$$

term3

$$- \left(\frac{1}{4} * a_t * \hat{\mu} - \frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \frac{1}{2} * \frac{a_t * \hat{\mu}}{\varepsilon} \right) * M_s^{-2} \varepsilon_{i,j} * \varepsilon^{\alpha_1, \alpha_4} * \varepsilon^{\alpha_3, \alpha_2} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_4} * \varepsilon_{\dot{\alpha}_3, \dot{\alpha}_2} * (D_{\dot{\alpha}_1}^{\dot{\alpha}_1} H)_i * (D_{\dot{\alpha}_2}^{\dot{\alpha}_2} D_{\dot{\alpha}_3}^{\dot{\alpha}_3} D_{\dot{\alpha}_4}^{\dot{\alpha}_4} H^\dagger)_j \quad (2)$$

term4

$$- \left(\frac{1}{4} * a_t * \hat{\mu} - \frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \frac{1}{2} * \frac{a_t * \hat{\mu}}{\varepsilon} \right) * M_s^{-2} \varepsilon_{i,j} * \varepsilon^{\alpha_3, \alpha_2} * \varepsilon^{\alpha_1, \alpha_4} * \varepsilon_{\dot{\alpha}_3, \dot{\alpha}_2} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_4} * (D_{\dot{\alpha}_1}^{\dot{\alpha}_1} D_{\dot{\alpha}_2}^{\dot{\alpha}_2} D_{\dot{\alpha}_3}^{\dot{\alpha}_3} H)_i * (D_{\dot{\alpha}_4}^{\dot{\alpha}_4} H^\dagger)_j \quad (3)$$

term5

$$- \left(\frac{1}{2} * a_t * \hat{\mu} - \frac{1}{2} * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \frac{a_t * \hat{\mu}}{\varepsilon} \right) * M_s^{-2} \varepsilon_{i,j} * \varepsilon^{\alpha_2, \alpha_4} * \varepsilon^{\alpha_1, \alpha_3} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_4} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_3} * (D_{\dot{\alpha}_1}^{\dot{\alpha}_1} D_{\dot{\alpha}_2}^{\dot{\alpha}_2} H)_i * (D_{\dot{\alpha}_3}^{\dot{\alpha}_3} D_{\dot{\alpha}_4}^{\dot{\alpha}_4} H^\dagger)_j \quad (4)$$

term6

$$- \left(\frac{1}{8} * a_t * \hat{\mu} - \frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \frac{1}{4} * \frac{a_t * \hat{\mu}}{\varepsilon} \right) * M_s^{-2} \varepsilon_{i,j} * \varepsilon^{\alpha_4, \alpha_3} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_4, \dot{\alpha}_3} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * (D_{\dot{\alpha}_1}^{\dot{\alpha}_1} D_{\dot{\alpha}_2}^{\dot{\alpha}_2} D_{\dot{\alpha}_3}^{\dot{\alpha}_3} D_{\dot{\alpha}_4}^{\dot{\alpha}_4} H)_i * (H^\dagger)_j \quad (5)$$

term7

$$- \left(\frac{1}{4} * a_t * \hat{\mu} - \frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \frac{1}{2} * \frac{a_t * \hat{\mu}}{\varepsilon} \right) * M_s^{-2} \varepsilon_{i,j} * \varepsilon^{\alpha_3, \alpha_4} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_3, \dot{\alpha}_4} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * (D_{\dot{\alpha}_1}^{\dot{\alpha}_1} D_{\dot{\alpha}_2}^{\dot{\alpha}_2} D_{\dot{\alpha}_3}^{\dot{\alpha}_3} H)_i * (D_{\dot{\alpha}_4}^{\dot{\alpha}_4} H^\dagger)_j \quad (6)$$

term8

$$- \left(\frac{1}{4} * a_t * \hat{\mu} - \frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \frac{1}{2} * \frac{a_t * \hat{\mu}}{\varepsilon} \right) * M_s^{-2} \varepsilon_{i,j} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon^{\alpha_4, \alpha_3} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * \varepsilon_{\dot{\alpha}_4, \dot{\alpha}_3} * (D_{\dot{\alpha}_1}^{\dot{\alpha}_1} H)_i * (D_{\dot{\alpha}_2}^{\dot{\alpha}_2} D_{\dot{\alpha}_3}^{\dot{\alpha}_3} D_{\dot{\alpha}_4}^{\dot{\alpha}_4} H^\dagger)_j \quad (7)$$

term10

$$- \left(\frac{1}{8} * a_t * \hat{\mu} - \frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \frac{1}{4} * \frac{a_t * \hat{\mu}}{\varepsilon} \right) * M_s^{-2} \varepsilon_{i,j} * \varepsilon^{\alpha_4, \alpha_3} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_4, \dot{\alpha}_3} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * H_i * (D_{\dot{\alpha}_1}^{\dot{\alpha}_1} D_{\dot{\alpha}_2}^{\dot{\alpha}_2} D_{\dot{\alpha}_3}^{\dot{\alpha}_3} D_{\dot{\alpha}_4}^{\dot{\alpha}_4} H^\dagger)_j \quad (8)$$

nD: 2, H: 1, H+: 1

term1

$$\begin{aligned}
& - \left(\frac{1}{4} * a_t^2 + \frac{1}{2} * a_t * \hat{\mu} - \frac{1}{2} * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} - \frac{1}{2} \right. \\
& \quad \left. * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 + \frac{5}{4} * a_t^2 * \hat{\mu}_h^2 + \frac{a_t * \hat{\mu}}{\varepsilon} + \frac{a_t^2 * \hat{\mu}_h^2}{\varepsilon} \right) \\
& \quad \varepsilon_{i,j} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (D_{\alpha_1}^{\dot{\alpha}_1} H)_i * (D_{\alpha_2}^{\dot{\alpha}_2} H^\dagger)_j
\end{aligned} \tag{9}$$

term2

$$\begin{aligned}
& - \left(\frac{1}{4} * a_t * \hat{\mu} - \frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \frac{1}{2} * \frac{a_t * \hat{\mu}}{\varepsilon} \right) \\
& \quad \varepsilon_{i,j} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * (D_{\alpha_1}^{\dot{\alpha}_1} D_{\alpha_2}^{\dot{\alpha}_2} H)_i * (H^\dagger)_j
\end{aligned} \tag{10}$$

term9

$$\begin{aligned}
& - \left(\frac{1}{4} * a_t * \hat{\mu} - \frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \frac{1}{2} * \frac{a_t * \hat{\mu}}{\varepsilon} \right) \\
& \quad \varepsilon_{i,j} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * H_i * (D_{\alpha_1}^{\dot{\alpha}_1} D_{\alpha_2}^{\dot{\alpha}_2} H^\dagger)_j
\end{aligned} \tag{11}$$

nD: 0, H: 1, H+: 1

term11

$$\begin{aligned}
& + \left(a_t^2 + a_t^2 * \hat{\mu}_h^2 + a_t^2 * \hat{\mu}_h^4 - \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 - \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^4 - \frac{1}{2} * a_t * \hat{\mu} + \frac{1}{2} \right. \\
& \quad \left. * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \frac{1}{2} * \kappa - \frac{1}{2} * \kappa * \ln \frac{M_s^2}{\mu_M^2} + 2 * \frac{a_t^2}{\varepsilon} + 2 * \frac{a_t^2 * \hat{\mu}_h^2}{\varepsilon} + 2 * \frac{a_t^2 * \hat{\mu}_h^4}{\varepsilon} - \frac{a_t * \hat{\mu}}{\varepsilon} + \frac{\kappa}{\varepsilon} \right) \\
& \quad * M_s^2 \\
& \quad \varepsilon_{i,j} * H_i * (H^\dagger)_j
\end{aligned} \tag{12}$$

nD: 2, BL: 1, H: 1, H+: 1

term12

$$\begin{aligned}
& - \left(\frac{1}{96} * g_1 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_1 * a_t^2}{\varepsilon} - \frac{17}{576} * g_1 * a_t^2 \right) * M_s^{-2} \\
& \quad \varepsilon_{i,j} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * \varepsilon^{\alpha_1, \alpha_3} * \varepsilon^{\alpha_2, \alpha_4} * (B_L)_{\alpha_1, \alpha_2} * (D_{\alpha_3}^{\dot{\alpha}_1} H)_i * (D_{\alpha_4}^{\dot{\alpha}_2} H^\dagger)_j
\end{aligned} \tag{13}$$

term12

$$\begin{aligned}
& - \left(\frac{1}{96} * g_1 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_1 * a_t^2}{\varepsilon} - \frac{17}{576} * g_1 * a_t^2 \right) * M_s^{-2} \\
& \quad \varepsilon_{i,j} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * \varepsilon^{\alpha_1, \alpha_4} * \varepsilon^{\alpha_2, \alpha_3} * (B_L)_{\alpha_1, \alpha_2} * (D_{\alpha_3}^{\dot{\alpha}_1} H)_i * (D_{\alpha_4}^{\dot{\alpha}_2} H^\dagger)_j
\end{aligned} \tag{14}$$

term13

$$\begin{aligned}
& - \left(\frac{1}{96} * g_1 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_1 * a_t^2}{\varepsilon} - \frac{17}{576} * g_1 * a_t^2 \right) * M_s^{-2} \\
& \quad \varepsilon_{i,j} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * \varepsilon^{\alpha_1, \alpha_3} * \varepsilon^{\alpha_2, \alpha_4} * (B_L)_{\alpha_1, \alpha_2} * (D_{\alpha_3}^{\dot{\alpha}_1} H)_i * (D_{\alpha_4}^{\dot{\alpha}_2} H^\dagger)_j
\end{aligned} \tag{15}$$

term13

$$- \left(\frac{1}{96} * g_1 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_1 * a_t^2}{\varepsilon} - \frac{17}{576} * g_1 * a_t^2 \right) * M_s^{-2} \quad (16)$$

$$\varepsilon_{i,j} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * \varepsilon^{\alpha_1, \alpha_4} * \varepsilon^{\alpha_2, \alpha_3} * (B_L)_{\alpha_1, \alpha_2} * (D_{\alpha_3}^{\dot{\alpha}_1} H)_i * (D_{\alpha_4}^{\dot{\alpha}_2} H^\dagger)_j$$

nD: 2, H: 1, H+: 1, BL+: 1

term12

$$+ \left(\frac{1}{96} * g_1 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_1 * a_t^2}{\varepsilon} - \frac{17}{576} * g_1 * a_t^2 \right) * M_s^{-2} \quad (17)$$

$$\varepsilon_{i,j} * \varepsilon^{\alpha_3, \alpha_4} * \varepsilon_{\dot{\alpha}_3, \dot{\alpha}_1} * \varepsilon_{\dot{\alpha}_4, \dot{\alpha}_2} * (D_{\alpha_3}^{\dot{\alpha}_1} H)_i * (D_{\alpha_4}^{\dot{\alpha}_2} H^\dagger)_j * (B_R)^{\dot{\alpha}_3, \dot{\alpha}_4}$$

term12

$$+ \left(\frac{1}{96} * g_1 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_1 * a_t^2}{\varepsilon} - \frac{17}{576} * g_1 * a_t^2 \right) * M_s^{-2} \quad (18)$$

$$\varepsilon_{i,j} * \varepsilon^{\alpha_3, \alpha_4} * \varepsilon_{\dot{\alpha}_3, \dot{\alpha}_2} * \varepsilon_{\dot{\alpha}_4, \dot{\alpha}_1} * (D_{\alpha_3}^{\dot{\alpha}_1} H)_i * (D_{\alpha_4}^{\dot{\alpha}_2} H^\dagger)_j * (B_R)^{\dot{\alpha}_3, \dot{\alpha}_4}$$

term13

$$+ \left(\frac{1}{96} * g_1 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_1 * a_t^2}{\varepsilon} - \frac{17}{576} * g_1 * a_t^2 \right) * M_s^{-2} \quad (19)$$

$$\varepsilon_{i,j} * \varepsilon^{\alpha_3, \alpha_4} * \varepsilon_{\dot{\alpha}_3, \dot{\alpha}_1} * \varepsilon_{\dot{\alpha}_4, \dot{\alpha}_2} * (D_{\alpha_3}^{\dot{\alpha}_1} H)_i * (D_{\alpha_4}^{\dot{\alpha}_2} H^\dagger)_j * (B_R)^{\dot{\alpha}_3, \dot{\alpha}_4}$$

term13

$$+ \left(\frac{1}{96} * g_1 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_1 * a_t^2}{\varepsilon} - \frac{17}{576} * g_1 * a_t^2 \right) * M_s^{-2} \quad (20)$$

$$\varepsilon_{i,j} * \varepsilon^{\alpha_3, \alpha_4} * \varepsilon_{\dot{\alpha}_3, \dot{\alpha}_2} * \varepsilon_{\dot{\alpha}_4, \dot{\alpha}_1} * (D_{\alpha_3}^{\dot{\alpha}_1} H)_i * (D_{\alpha_4}^{\dot{\alpha}_2} H^\dagger)_j * (B_R)^{\dot{\alpha}_3, \dot{\alpha}_4}$$

nD: 2, H: 2, H+: 2

term14

$$+ \left(a_t^3 * \hat{\mu} * \hat{\mu}_h^2 - a_t^4 * \hat{\mu}_h^2 + \ln \frac{M_s^2}{\mu_M^2} * a_t^4 * \hat{\mu}_h^2 - \kappa * a_t^2 - \kappa * a_t^2 * \hat{\mu}_h^2 - \kappa * a_t^2 * \hat{\mu}_h^4 + \kappa \right. \\ * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 + \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^4 + \frac{3}{32} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{8} \\ * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}^2 + \frac{1}{8} * \lambda_\varphi * a_t^2 - \frac{1}{8} * \lambda_\varphi * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{9}{64} * g_2^2 * a_t^2 - \frac{3}{16} * \frac{g_2^2 * a_t^2}{\varepsilon} - \frac{1}{4} \\ * a_t^2 * \hat{\mu}^2 - \frac{1}{4} * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} - \frac{1}{4} * \frac{a_t^2 * \hat{\mu}^2}{\varepsilon} + \frac{1}{4} * \frac{\lambda_\varphi * a_t^2}{\varepsilon} + \frac{1}{2} * a_t^3 * \hat{\mu} - \frac{1}{2} * a_t^4 - \frac{1}{2} \\ * \ln \frac{M_s^2}{\mu_M^2} * a_t^3 * \hat{\mu} * \hat{\mu}_h^2 + \frac{1}{2} * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 + \frac{1}{2} * \kappa * a_t * \hat{\mu} + \frac{1}{2} * \frac{\kappa * a_t * \hat{\mu}}{\varepsilon} - 2 * \frac{a_t^4 * \hat{\mu}_h^2}{\varepsilon} \\ \left. - 2 * \frac{\kappa * a_t^2}{\varepsilon} - 2 * \frac{\kappa * a_t^2 * \hat{\mu}_h^2}{\varepsilon} - 2 * \frac{\kappa * a_t^2 * \hat{\mu}_h^4}{\varepsilon} + \frac{a_t^3 * \hat{\mu} * \hat{\mu}_h^2}{\varepsilon} - \frac{a_t^4}{\varepsilon} \right) * M_s^{-2} \quad (21)$$

$$\varepsilon_{i,l} * \varepsilon_{j,k} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * (D_{\alpha_1}^{\dot{\alpha}_1} D_{\alpha_2}^{\dot{\alpha}_2} H)_i * H_j * (H^\dagger)_k * (H^\dagger)_l$$

term15

$$\begin{aligned}
& - \left(\frac{3}{32} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}^2 - \frac{1}{8} * \lambda_\varphi * a_t^2 + \frac{1}{8} * \lambda_\varphi * \ln \frac{M_s^2}{\mu_M^2} \right. \\
& * a_t^2 - \frac{9}{64} * g_2^2 * a_t^2 - \frac{3}{16} * \frac{g_2^2 * a_t^2}{\varepsilon} + \frac{1}{4} * a_t^2 * \hat{\mu}^2 - \frac{1}{4} * \kappa * a_t * \hat{\mu} + \frac{1}{4} * \kappa \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \frac{1}{4} * \frac{a_t^2 * \hat{\mu}^2}{\varepsilon} - \frac{1}{4} * \frac{\lambda_\varphi * a_t^2}{\varepsilon} - \frac{1}{2} * \frac{\kappa * a_t * \hat{\mu}}{\varepsilon} \Big) * M_s^{-2} \\
& \varepsilon_{i,k} * \varepsilon_{j,l} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * (D_{\alpha_1}^{\dot{\alpha}_1} D_{\alpha_2}^{\dot{\alpha}_2} H)_i * H_j * (H^\dagger)_k * (H^\dagger)_l
\end{aligned} \tag{22}$$

term16

$$\begin{aligned}
& + \left(a_t^3 * \hat{\mu} * \hat{\mu}_h^2 - a_t^4 * \hat{\mu}_h^2 + \ln \frac{M_s^2}{\mu_M^2} * a_t^4 * \hat{\mu}_h^2 - \kappa * a_t^2 - \kappa * a_t^2 * \hat{\mu}_h^2 - \kappa * a_t^2 * \hat{\mu}_h^4 + \kappa \right. \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 + \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^4 + \frac{3}{32} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{8} \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}^2 + \frac{1}{8} * \lambda_\varphi * a_t^2 - \frac{1}{8} * \lambda_\varphi * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{9}{64} * g_2^2 * a_t^2 - \frac{3}{16} * \frac{g_2^2 * a_t^2}{\varepsilon} - \frac{1}{4} \\
& * a_t^2 * \hat{\mu}^2 - \frac{1}{4} * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} - \frac{1}{4} * \frac{a_t^2 * \hat{\mu}^2}{\varepsilon} + \frac{1}{4} * \frac{\lambda_\varphi * a_t^2}{\varepsilon} + \frac{1}{2} * a_t^3 * \hat{\mu} - \frac{1}{2} * a_t^4 - \frac{1}{2} \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t^3 * \hat{\mu} * \hat{\mu}_h^2 + \frac{1}{2} * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 + \frac{1}{2} * \kappa * a_t * \hat{\mu} + \frac{1}{2} * \frac{\kappa * a_t * \hat{\mu}}{\varepsilon} - 2 * \frac{a_t^4 * \hat{\mu}_h^2}{\varepsilon} \\
& - 2 * \frac{\kappa * a_t^2}{\varepsilon} - 2 * \frac{\kappa * a_t^2 * \hat{\mu}_h^2}{\varepsilon} - 2 * \frac{\kappa * a_t^2 * \hat{\mu}_h^4}{\varepsilon} + \frac{a_t^3 * \hat{\mu} * \hat{\mu}_h^2}{\varepsilon} - \frac{a_t^4}{\varepsilon} \Big) * M_s^{-2} \\
& \varepsilon_{i,l} * \varepsilon_{j,k} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * H_i * H_j * (D_{\alpha_1}^{\dot{\alpha}_1} D_{\alpha_2}^{\dot{\alpha}_2} H^\dagger)_k * (H^\dagger)_l
\end{aligned} \tag{23}$$

term17

$$\begin{aligned}
& - \left(\frac{3}{32} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}^2 - \frac{1}{8} * \lambda_\varphi * a_t^2 + \frac{1}{8} * \lambda_\varphi * \ln \frac{M_s^2}{\mu_M^2} \right. \\
& * a_t^2 - \frac{9}{64} * g_2^2 * a_t^2 - \frac{3}{16} * \frac{g_2^2 * a_t^2}{\varepsilon} + \frac{1}{4} * a_t^2 * \hat{\mu}^2 - \frac{1}{4} * \kappa * a_t * \hat{\mu} + \frac{1}{4} * \kappa \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \frac{1}{4} * \frac{a_t^2 * \hat{\mu}^2}{\varepsilon} - \frac{1}{4} * \frac{\lambda_\varphi * a_t^2}{\varepsilon} - \frac{1}{2} * \frac{\kappa * a_t * \hat{\mu}}{\varepsilon} \Big) * M_s^{-2} \\
& \varepsilon_{i,k} * \varepsilon_{j,l} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * H_i * H_j * (D_{\alpha_1}^{\dot{\alpha}_1} D_{\alpha_2}^{\dot{\alpha}_2} H^\dagger)_k * (H^\dagger)_l
\end{aligned} \tag{24}$$

term18

$$\begin{aligned}
& - \left(a_t^2 * \hat{\mu}^2 + \ln \frac{M_s^2}{\mu_M^2} * a_t^3 * \hat{\mu} * \hat{\mu}_h^2 + \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} - \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 \right. \\
& + \frac{1}{24} * d * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{11}{144} * d * g_2^2 * a_t^2 - \frac{1}{12} * \frac{d * g_2^2 * a_t^2}{\varepsilon} - \frac{1}{8} * \varepsilon * \ln \frac{M_s^2}{\mu_M^2} \\
& * a_t^3 * \hat{\mu} + \frac{1}{8} * \varepsilon * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{4} * \varepsilon * a_t^3 * \hat{\mu} - \frac{1}{4} * \varepsilon * \kappa * a_t^2 - \frac{29}{72} * g_2^2 * a_t^2 \\
& - \frac{1}{2} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}^2 - \frac{1}{2} * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 - \frac{1}{2} * \kappa * a_t^2 * \hat{\mu}_h^2 - \frac{1}{2} * \lambda_\varphi * a_t^2 + \frac{1}{2} * \lambda_\varphi \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{2} * g_2^2 * a_t^2 * \hat{\mu}_h^2 + \frac{1}{2} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 - \frac{1}{2} * \lambda_h * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 \\
& + \frac{1}{2} * \varepsilon * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 + \frac{7}{12} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{4} * a_t^3 * \hat{\mu} - \frac{3}{4} * \varepsilon * a_t^4 - \frac{7}{6} * \frac{g_2^2 * a_t^2}{\varepsilon} \\
& - \frac{5}{4} * a_t^4 + \frac{5}{4} * \kappa * a_t^2 - \frac{3}{2} * \kappa * a_t * \hat{\mu} + \frac{9}{4} * \lambda_h * a_t^2 - 2 * a_t^3 * \hat{\mu} * \hat{\mu}_h^2 + 2 * a_t^4 * \hat{\mu}_h^2 - 2 \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 * \hat{\mu}_h^2 + 2 * \kappa * a_t^2 * \hat{\mu}_h^4 - 2 * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - 2 * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^4 - 2 \\
& * \frac{a_t^3 * \hat{\mu} * \hat{\mu}_h^2}{\varepsilon} - 2 * \frac{\kappa * a_t * \hat{\mu}}{\varepsilon} + 2 * \frac{\kappa * a_t^2 * \hat{\mu}_h^2}{\varepsilon} + 4 * \frac{a_t^4 * \hat{\mu}_h^2}{\varepsilon} + 4 * \frac{\kappa * a_t^2}{\varepsilon} + 4 \\
& * \frac{\kappa * a_t^2 * \hat{\mu}_h^4}{\varepsilon} + \frac{a_t^2 * \hat{\mu}^2}{\varepsilon} + \frac{a_t^4}{\varepsilon} - \frac{\lambda_\varphi * a_t^2}{\varepsilon} - \frac{g_2^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} + \frac{\lambda_h * a_t^2}{\varepsilon} \Big) * M_s^{-2} \\
& \varepsilon_{i,k} * \varepsilon_{j,l} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (D_{\alpha_1}^{\dot{\alpha}_1} H)_i * H_j * (D_{\alpha_2}^{\dot{\alpha}_2} H^\dagger)_k * (H^\dagger)_l
\end{aligned} \tag{25}$$

term19

$$\begin{aligned}
& + \left(\frac{1}{96} * d * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{11}{576} * d * g_1^2 * a_t^2 - \frac{1}{48} * \kappa^2 + \frac{1}{48} * d * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} \right. \\
& * a_t^2 - \frac{1}{48} * \frac{d * g_1^2 * a_t^2}{\varepsilon} - \frac{11}{288} * d * g_2^2 * a_t^2 - \frac{1}{24} * g_1^2 * a_t^2 - \frac{1}{24} * \frac{d * g_2^2 * a_t^2}{\varepsilon} \\
& - \frac{5}{96} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{5}{48} * \frac{g_2^2 * a_t^2}{\varepsilon} + \frac{1}{8} * g_2^2 * a_t^2 * \hat{\mu}_h^2 - \frac{1}{8} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 \\
& * \hat{\mu}_h^2 - \frac{1}{8} * g_1^2 * a_t^2 * \hat{\mu}_h^2 + \frac{1}{8} * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{8} * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 - \frac{1}{6} * a_t^3 \\
& * \hat{\mu} - \frac{3}{16} * \varepsilon * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 + \frac{121}{576} * g_2^2 * a_t^2 + \frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}^2 - \frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} \\
& * a_t^4 + \frac{1}{4} * \lambda_h * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{4} * \frac{g_2^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} - \frac{1}{4} * \frac{g_1^2 * a_t^2}{\varepsilon} - \frac{1}{4} * \frac{g_1^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} \\
& - \frac{13}{48} * a_t^2 * \hat{\mu}^2 + \frac{11}{32} * \varepsilon * a_t^4 + \frac{5}{12} * \kappa * a_t^2 - \frac{1}{2} * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} - \frac{1}{2} * \kappa \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 - \frac{1}{2} * \frac{a_t^2 * \hat{\mu}^2}{\varepsilon} + \frac{1}{2} * \frac{a_t^4}{\varepsilon} - \frac{1}{2} * \frac{\lambda_h * a_t^2}{\varepsilon} + \frac{13}{24} * \kappa * a_t * \hat{\mu} \\
& - \frac{7}{8} * \lambda_h * a_t^2 + \frac{7}{6} * a_t^4 + \frac{5}{4} * \kappa * a_t^2 * \hat{\mu}_h^2 + \frac{\kappa * a_t * \hat{\mu}}{\varepsilon} + \frac{\kappa * a_t^2 * \hat{\mu}_h^2}{\varepsilon} \Big) * M_s^{-2} \\
& \varepsilon_{i,k} * \varepsilon_{j,l} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (D_{\alpha_1}^{\dot{\alpha}_1} H)_i * (D_{\alpha_2}^{\dot{\alpha}_2} H)_j * (H^\dagger)_k * (H^\dagger)_l
\end{aligned} \tag{26}$$

term20

$$\begin{aligned}
& - \left(\kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} + \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 - \frac{1}{48} * d * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{48} * d * g_1^2 \right. \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{11}{288} * d * g_2^2 * a_t^2 - \frac{11}{288} * d * g_1^2 * a_t^2 + \frac{1}{24} * \kappa^2 + \frac{1}{24} * \frac{d * g_2^2 * a_t^2}{\varepsilon} - \frac{1}{24} \\
& * \frac{d * g_1^2 * a_t^2}{\varepsilon} + \frac{29}{144} * g_2^2 * a_t^2 - \frac{29}{144} * g_1^2 * a_t^2 + \frac{1}{4} * g_2^2 * a_t^2 * \hat{\mu}_h^2 - \frac{1}{4} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 \\
& * \hat{\mu}_h^2 - \frac{1}{4} * g_1^2 * a_t^2 * \hat{\mu}_h^2 + \frac{1}{4} * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 - \frac{7}{24} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{7}{24} * g_1^2 \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{3} * a_t^3 * \hat{\mu} + \frac{3}{8} * \varepsilon * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 - \frac{1}{2} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}^2 + \frac{1}{2} * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 \\
& - \frac{1}{2} * \lambda_h * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{2} * \frac{g_2^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} - \frac{1}{2} * \frac{g_1^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} + \frac{13}{24} * a_t^2 * \hat{\mu}^2 + \frac{7}{12} \\
& * \frac{g_2^2 * a_t^2}{\varepsilon} - \frac{7}{12} * \frac{g_1^2 * a_t^2}{\varepsilon} - \frac{11}{16} * \varepsilon * a_t^4 - \frac{5}{6} * \kappa * a_t^2 - \frac{13}{12} * \kappa * a_t * \hat{\mu} + \frac{9}{4} * \lambda_h * a_t^2 - \frac{7}{3} \\
& * a_t^4 - \frac{5}{2} * \kappa * a_t^2 * \hat{\mu}_h^2 - 2 * \frac{\kappa * a_t * \hat{\mu}}{\varepsilon} - 2 * \frac{\kappa * a_t^2 * \hat{\mu}_h^2}{\varepsilon} + \frac{a_t^2 * \hat{\mu}^2}{\varepsilon} - \frac{a_t^4}{\varepsilon} + \frac{\lambda_h * a_t^2}{\varepsilon} \Big) \\
& * M_s^{-2} \\
& \varepsilon_{i,l} * \varepsilon_{j,k} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * (D_{\alpha_1}^{\dot{\alpha}_1} H)_i * H_j * (D_{\alpha_2}^{\dot{\alpha}_2} H^\dagger)_k * (H^\dagger)_l
\end{aligned} \tag{27}$$

term21

$$\begin{aligned}
& - \left(\frac{1}{96} * d * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{11}{576} * d * g_2^2 * a_t^2 - \frac{1}{48} * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{d * g_2^2 * a_t^2}{\varepsilon} \right. \\
& + \frac{1}{24} * \frac{g_1^2 * a_t^2}{\varepsilon} + \frac{17}{288} * g_1^2 * a_t^2 - \frac{19}{96} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{4} * g_2^2 * a_t^2 * \hat{\mu}_h^2 - \frac{1}{4} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} \\
& * a_t^2 * \hat{\mu}_h^2 + \frac{1}{4} * \lambda_h * a_t^2 + \frac{179}{576} * g_2^2 * a_t^2 + \frac{19}{48} * \frac{g_2^2 * a_t^2}{\varepsilon} + \frac{1}{2} * \frac{g_2^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} \Big) * M_s^{-2} \\
& \varepsilon_{i,l} * \varepsilon_{j,k} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (D_{\alpha_1}^{\dot{\alpha}_1} H)_i * (D_{\alpha_2}^{\dot{\alpha}_2} H)_j * (H^\dagger)_k * (H^\dagger)_l
\end{aligned} \tag{28}$$

term23

$$\begin{aligned}
& + \left(\frac{1}{96} * d * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{11}{576} * d * g_1^2 * a_t^2 - \frac{1}{48} * \kappa^2 + \frac{1}{48} * d * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} \right. \\
& * a_t^2 - \frac{1}{48} * \frac{d * g_1^2 * a_t^2}{\varepsilon} - \frac{11}{288} * d * g_2^2 * a_t^2 - \frac{1}{24} * g_1^2 * a_t^2 - \frac{1}{24} * \frac{d * g_2^2 * a_t^2}{\varepsilon} \\
& - \frac{5}{96} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{5}{48} * \frac{g_2^2 * a_t^2}{\varepsilon} + \frac{1}{8} * g_2^2 * a_t^2 * \hat{\mu}_h^2 - \frac{1}{8} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 \\
& * \hat{\mu}_h^2 - \frac{1}{8} * g_1^2 * a_t^2 * \hat{\mu}_h^2 + \frac{1}{8} * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{8} * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 - \frac{1}{6} * a_t^3 \\
& * \hat{\mu} - \frac{3}{16} * \varepsilon * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 + \frac{121}{576} * g_2^2 * a_t^2 + \frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}^2 - \frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} \\
& * a_t^4 + \frac{1}{4} * \lambda_h * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{4} * \frac{g_2^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} - \frac{1}{4} * \frac{g_1^2 * a_t^2}{\varepsilon} - \frac{1}{4} * \frac{g_1^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} \\
& - \frac{13}{48} * a_t^2 * \hat{\mu}^2 + \frac{11}{32} * \varepsilon * a_t^4 + \frac{5}{12} * \kappa * a_t^2 - \frac{1}{2} * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t * \hat{\mu} - \frac{1}{2} * \kappa \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 - \frac{1}{2} * \frac{a_t^2 * \hat{\mu}^2}{\varepsilon} + \frac{1}{2} * \frac{a_t^4}{\varepsilon} - \frac{1}{2} * \frac{\lambda_h * a_t^2}{\varepsilon} + \frac{13}{24} * \kappa * a_t * \hat{\mu} \\
& - \frac{7}{8} * \lambda_h * a_t^2 + \frac{7}{6} * a_t^4 + \frac{5}{4} * \kappa * a_t^2 * \hat{\mu}_h^2 + \frac{\kappa * a_t * \hat{\mu}}{\varepsilon} + \frac{\kappa * a_t^2 * \hat{\mu}_h^2}{\varepsilon} \Big) * M_s^{-2} \\
& \varepsilon_{i,k} * \varepsilon_{j,l} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * H_i * H_j * (D_{\alpha_1}^{\dot{\alpha}_1} H^\dagger)_k * (D_{\alpha_2}^{\dot{\alpha}_2} H^\dagger)_l
\end{aligned} \tag{29}$$

term24

$$\begin{aligned}
& - \left(\frac{1}{96} * d * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{11}{576} * d * g_2^2 * a_t^2 - \frac{1}{48} * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{d * g_2^2 * a_t^2}{\varepsilon} \right. \\
& + \frac{1}{24} * \frac{g_1^2 * a_t^2}{\varepsilon} + \frac{17}{288} * g_1^2 * a_t^2 - \frac{19}{96} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{4} * g_2^2 * a_t^2 * \hat{\mu}_h^2 - \frac{1}{4} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} \\
& * a_t^2 * \hat{\mu}_h^2 + \frac{1}{4} * \lambda_h * a_t^2 + \frac{179}{576} * g_2^2 * a_t^2 + \frac{19}{48} * \frac{g_2^2 * a_t^2}{\varepsilon} + \frac{1}{2} * \frac{g_2^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} \Big) * M_s^{-2} \\
& \varepsilon_{i,l} * \varepsilon_{j,k} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * H_i * H_j * (D_{\alpha_1}^{\dot{\alpha}_1} H^\dagger)_k * (D_{\alpha_2}^{\dot{\alpha}_2} H^\dagger)_l
\end{aligned} \tag{30}$$

nD: 0, H: 2, H+: 2

term26

$$\begin{aligned}
& + \left(a_t^3 * \hat{\mu} - \ln \frac{M_s^2}{\mu_M^2} * a_t^3 * \hat{\mu} * \hat{\mu}_h^2 - \frac{1}{4} * a_t^2 * \hat{\mu}^2 - \frac{1}{4} * \kappa^2 * \ln \frac{M_s^2}{\mu_M^2} + \frac{1}{4} * \lambda_\varphi * a_t^2 \right. \\
& - \frac{1}{4} * \lambda_\varphi * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{2} * \kappa * a_t * \hat{\mu} + \frac{1}{2} * \frac{\kappa^2}{\varepsilon} + \frac{1}{2} * \frac{\lambda_\varphi * a_t^2}{\varepsilon} + 2 * a_t^3 * \hat{\mu} * \hat{\mu}_h^2 \\
& + 2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 - 2 * \kappa * a_t^2 * \hat{\mu}_h^4 + 2 * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + 2 * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^4 \\
& + 2 * \frac{a_t^3 * \hat{\mu} * \hat{\mu}_h^2}{\varepsilon} - 3 * a_t^4 - 3 * \kappa * a_t^2 + 3 * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 + 3 * \lambda_h * a_t^2 \\
& - 3 * \lambda_h * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - 4 * \kappa * a_t^2 * \hat{\mu}_h^2 - 4 * \frac{a_t^4}{\varepsilon} - 4 * \frac{\kappa * a_t^2}{\varepsilon} - 4 * \frac{\kappa * a_t^2 * \hat{\mu}_h^4}{\varepsilon} \\
& + 6 * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 * \hat{\mu}_h^2 + 6 * \lambda_h * a_t^2 * \hat{\mu}_h^2 - 6 * \lambda_h * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 - 6 \\
& * \frac{\kappa * a_t^2 * \hat{\mu}_h^2}{\varepsilon} + 6 * \frac{\lambda_h * a_t^2}{\varepsilon} - 8 * a_t^4 * \hat{\mu}_h^2 - 12 * \frac{a_t^4 * \hat{\mu}_h^2}{\varepsilon} + 12 * \frac{\lambda_h * a_t^2 * \hat{\mu}_h^2}{\varepsilon} \Big) \\
& \varepsilon_{i,k} * \varepsilon_{j,l} * H_i * H_j * (H^\dagger)_k * (H^\dagger)_l
\end{aligned} \tag{31}$$

nD: 2, WL: 1, H: 1, H+: 1

term22

$$\begin{aligned}
& - \left(\frac{1}{96} * g_2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_2 * a_t^2}{\varepsilon} - \frac{17}{576} * g_2 * a_t^2 \right) * M_s^{-2} \\
& \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * \varepsilon^{\alpha_1, \alpha_4} * \varepsilon^{\alpha_2, \alpha_3} * \delta_{k,i} * \delta_{l,j} * (W_L)_{i,j, \alpha_1, \alpha_2} * (D_{\alpha_3}^{\dot{\alpha}_1} H)_k * (D_{\alpha_4}^{\dot{\alpha}_2} H^\dagger)_l
\end{aligned} \tag{32}$$

term22

$$\begin{aligned}
& - \left(\frac{1}{96} * g_2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_2 * a_t^2}{\varepsilon} - \frac{17}{576} * g_2 * a_t^2 \right) * M_s^{-2} \\
& \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * \varepsilon^{\alpha_1, \alpha_3} * \varepsilon^{\alpha_2, \alpha_4} * \delta_{k,i} * \delta_{l,j} * (W_L)_{i,j, \alpha_1, \alpha_2} * (D_{\alpha_3}^{\dot{\alpha}_1} H)_k * (D_{\alpha_4}^{\dot{\alpha}_2} H^\dagger)_l
\end{aligned} \tag{33}$$

term25

$$\begin{aligned}
& - \left(\frac{1}{96} * g_2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_2 * a_t^2}{\varepsilon} - \frac{17}{576} * g_2 * a_t^2 \right) * M_s^{-2} \\
& \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * \varepsilon^{\alpha_1, \alpha_4} * \varepsilon^{\alpha_2, \alpha_3} * \delta_{k,i} * \delta_{l,j} * (W_L)_{i,j, \alpha_1, \alpha_2} * (D_{\alpha_3}^{\dot{\alpha}_1} H)_k * (D_{\alpha_4}^{\dot{\alpha}_2} H^\dagger)_l
\end{aligned} \tag{34}$$

term25

$$\begin{aligned}
& - \left(\frac{1}{96} * g_2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_2 * a_t^2}{\varepsilon} - \frac{17}{576} * g_2 * a_t^2 \right) * M_s^{-2} \\
& \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * \varepsilon^{\alpha_1, \alpha_3} * \varepsilon^{\alpha_2, \alpha_4} * \delta_{k,i} * \delta_{l,j} * (W_L)_{i,j, \alpha_1, \alpha_2} * (D_{\alpha_3}^{\dot{\alpha}_1} H)_k * (D_{\alpha_4}^{\dot{\alpha}_2} H^\dagger)_l
\end{aligned} \tag{35}$$

nD: 2, H: 1, H+: 1, WL+: 1

term22

$$+ \left(\frac{1}{96} * g_2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_2 * a_t^2}{\varepsilon} - \frac{17}{576} * g_2 * a_t^2 \right) * M_s^{-2} \quad (36)$$

$$\varepsilon^{\alpha_4, \alpha_3} * \varepsilon_{\dot{\alpha}_3, \dot{\alpha}_2} * \varepsilon_{\dot{\alpha}_4, \dot{\alpha}_1} * \delta_{k,i} * \delta_{l,j} * (D_{\alpha_3}^{\dot{\alpha}_1} H)_k * (D_{\alpha_4}^{\dot{\alpha}_2} H^\dagger)_l * (W_R)_{i,j}^{\dot{\alpha}_3, \dot{\alpha}_4}$$

term22

$$+ \left(\frac{1}{96} * g_2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_2 * a_t^2}{\varepsilon} - \frac{17}{576} * g_2 * a_t^2 \right) * M_s^{-2} \quad (37)$$

$$\varepsilon^{\alpha_4, \alpha_3} * \varepsilon_{\dot{\alpha}_3, \dot{\alpha}_1} * \varepsilon_{\dot{\alpha}_4, \dot{\alpha}_2} * \delta_{k,i} * \delta_{l,j} * (D_{\alpha_3}^{\dot{\alpha}_1} H)_k * (D_{\alpha_4}^{\dot{\alpha}_2} H^\dagger)_l * (W_R)_{i,j}^{\dot{\alpha}_3, \dot{\alpha}_4}$$

term25

$$+ \left(\frac{1}{96} * g_2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_2 * a_t^2}{\varepsilon} - \frac{17}{576} * g_2 * a_t^2 \right) * M_s^{-2} \quad (38)$$

$$\varepsilon^{\alpha_4, \alpha_3} * \varepsilon_{\dot{\alpha}_3, \dot{\alpha}_2} * \varepsilon_{\dot{\alpha}_4, \dot{\alpha}_1} * \delta_{k,i} * \delta_{l,j} * (D_{\alpha_3}^{\dot{\alpha}_1} H)_k * (D_{\alpha_4}^{\dot{\alpha}_2} H^\dagger)_l * (W_R)_{i,j}^{\dot{\alpha}_3, \dot{\alpha}_4}$$

term25

$$+ \left(\frac{1}{96} * g_2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{48} * \frac{g_2 * a_t^2}{\varepsilon} - \frac{17}{576} * g_2 * a_t^2 \right) * M_s^{-2} \quad (39)$$

$$\varepsilon^{\alpha_4, \alpha_3} * \varepsilon_{\dot{\alpha}_3, \dot{\alpha}_1} * \varepsilon_{\dot{\alpha}_4, \dot{\alpha}_2} * \delta_{k,i} * \delta_{l,j} * (D_{\alpha_3}^{\dot{\alpha}_1} H)_k * (D_{\alpha_4}^{\dot{\alpha}_2} H^\dagger)_l * (W_R)_{i,j}^{\dot{\alpha}_3, \dot{\alpha}_4}$$

nD: 0, H: 1, H+: 1, BL+: 2

term27

$$- \left(\frac{5}{576} * g_1^2 * a_t^2 - \frac{1}{96} * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{48} * \frac{g_1^2 * a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (40)$$

$$\varepsilon_{i,j} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_3} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_4} * H_i * (H^\dagger)_j * (B_R)^{\dot{\alpha}_1, \dot{\alpha}_2} * (B_R)^{\dot{\alpha}_3, \dot{\alpha}_4}$$

term27

$$- \left(\frac{5}{576} * g_1^2 * a_t^2 - \frac{1}{96} * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{48} * \frac{g_1^2 * a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (41)$$

$$\varepsilon_{i,j} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_4} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_3} * H_i * (H^\dagger)_j * (B_R)^{\dot{\alpha}_1, \dot{\alpha}_2} * (B_R)^{\dot{\alpha}_3, \dot{\alpha}_4}$$

nD: 0, BL: 2, H: 1, H+: 1

term27

$$- \left(\frac{5}{576} * g_1^2 * a_t^2 - \frac{1}{96} * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{48} * \frac{g_1^2 * a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (42)$$

$$\varepsilon_{i,j} * \varepsilon^{\alpha_1, \alpha_3} * \varepsilon^{\alpha_2, \alpha_4} * (B_L)_{\alpha_1, \alpha_2} * (B_L)_{\alpha_3, \alpha_4} * H_i * (H^\dagger)_j$$

term27

$$- \left(\frac{5}{576} * g_1^2 * a_t^2 - \frac{1}{96} * g_1^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{48} * \frac{g_1^2 * a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (43)$$

$$\varepsilon_{i,j} * \varepsilon^{\alpha_1, \alpha_4} * \varepsilon^{\alpha_2, \alpha_3} * (B_L)_{\alpha_1, \alpha_2} * (B_L)_{\alpha_3, \alpha_4} * H_i * (H^\dagger)_j$$

nD: 0, H: 1, H+: 1, BL+: 1, WL+: 1

term28

$$+ \left(\frac{5}{288} * g_1 * g_2 * a_t^2 - \frac{1}{48} * g_1 * g_2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{24} * \frac{g_1 * g_2 * a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (44)$$

$$\varepsilon^{\dot{\alpha}_1, \dot{\alpha}_3} * \varepsilon^{\dot{\alpha}_2, \dot{\alpha}_4} * \delta_{i,k} * \delta_{j,l} * H_i * (H^\dagger)_j * (B_R)^{\dot{\alpha}_1, \dot{\alpha}_2} * (W_R)^{\dot{\alpha}_3, \dot{\alpha}_4}_{k,l}$$

term28

$$+ \left(\frac{5}{288} * g_1 * g_2 * a_t^2 - \frac{1}{48} * g_1 * g_2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{24} * \frac{g_1 * g_2 * a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (45)$$

$$\varepsilon^{\dot{\alpha}_1, \dot{\alpha}_4} * \varepsilon^{\dot{\alpha}_2, \dot{\alpha}_3} * \delta_{i,k} * \delta_{j,l} * H_i * (H^\dagger)_j * (B_R)^{\dot{\alpha}_1, \dot{\alpha}_2} * (W_R)^{\dot{\alpha}_3, \dot{\alpha}_4}_{k,l}$$

nD: 0, BL: 1, WL: 1, H: 1, H+: 1

term28

$$+ \left(\frac{5}{288} * g_1 * g_2 * a_t^2 - \frac{1}{48} * g_1 * g_2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{24} * \frac{g_1 * g_2 * a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (46)$$

$$\varepsilon^{\alpha_1, \alpha_3} * \varepsilon^{\alpha_2, \alpha_4} * \delta_{i,k} * \delta_{j,l} * (B_L)_{\alpha_1, \alpha_2} * (W_L)_{k,l, \alpha_3, \alpha_4} * H_i * (H^\dagger)_j$$

term28

$$+ \left(\frac{5}{288} * g_1 * g_2 * a_t^2 - \frac{1}{48} * g_1 * g_2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{24} * \frac{g_1 * g_2 * a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (47)$$

$$\varepsilon^{\alpha_1, \alpha_4} * \varepsilon^{\alpha_2, \alpha_3} * \delta_{i,k} * \delta_{j,l} * (B_L)_{\alpha_1, \alpha_2} * (W_L)_{k,l, \alpha_3, \alpha_4} * H_i * (H^\dagger)_j$$

nD: 0, H: 3, H+: 3

term29

$$\begin{aligned}
& + \left(a_t^4 * \hat{\mu}^2 - \kappa * a_t^3 * \hat{\mu} * \hat{\mu}_h^4 + \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^3 * \hat{\mu} + \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^3 \right. \\
& * \hat{\mu} * \hat{\mu}_h^4 - \lambda_\varphi * a_t^4 * \hat{\mu}_h^2 - \frac{1}{12} * \kappa^3 - \frac{1}{12} * \lambda_\varphi * \ln \frac{M_s^2}{\mu_M^2} * a_t^3 * \hat{\mu} - \frac{1}{6} * a_t^3 \\
& * \hat{\mu}^3 + \frac{1}{6} * \frac{\lambda_\varphi * a_t^3 * \hat{\mu}}{\varepsilon} - \frac{1}{4} * \kappa^2 * a_t * \hat{\mu} + \frac{1}{4} * \lambda_\varphi * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 \\
& + \frac{1}{3} * \lambda_\varphi * a_t^3 * \hat{\mu} + \frac{1}{2} * \kappa * a_t^2 * \hat{\mu}^2 - \frac{1}{2} * \lambda_\varphi * a_t^4 + \frac{1}{2} * \lambda_\varphi * \ln \frac{M_s^2}{\mu_M^2} \\
& * a_t^4 * \hat{\mu}_h^2 - \frac{1}{2} * \lambda_\varphi * \kappa * a_t^2 - \frac{1}{2} * \frac{\lambda_\varphi * \kappa * a_t^2}{\varepsilon} - \frac{3}{2} * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 * \hat{\mu}^2 \\
& * \hat{\mu}_h^2 + \frac{7}{2} * a_t^4 * \hat{\mu}^2 * \hat{\mu}_h^2 + \frac{7}{2} * \ln \frac{M_s^2}{\mu_M^2} * a_t^5 * \hat{\mu} + \frac{11}{2} * \kappa^2 * a_t^2 - \frac{15}{2} * a_t^5 \\
& * \hat{\mu} + \frac{19}{2} * \kappa^2 * a_t^2 * \hat{\mu}_h^2 - \frac{21}{2} * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 + \frac{43}{3} * a_t^6 + \frac{37}{2} * \kappa * a_t^4 \\
& - 2 * \lambda_h * \ln \frac{M_s^2}{\mu_M^2} * a_t^3 * \hat{\mu} - 2 * \frac{\kappa * a_t^3 * \hat{\mu}}{\varepsilon} - 2 * \frac{\kappa * a_t^3 * \hat{\mu} * \hat{\mu}_h^4}{\varepsilon} + 3 \\
& * a_t^6 * \hat{\mu}_h^2 - 3 * \ln \frac{M_s^2}{\mu_M^2} * a_t^6 * \hat{\mu}_h^2 + 3 * \kappa^2 * a_t^2 * \hat{\mu}_h^4 - 3 * \kappa^2 * \ln \frac{M_s^2}{\mu_M^2} \\
& * a_t^2 - 3 * \kappa^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^4 + 3 * \frac{a_t^4 * \hat{\mu}^2 * \hat{\mu}_h^2}{\varepsilon} + 4 * \lambda_h * a_t^3 * \hat{\mu} \\
& + 4 * \lambda_h^2 * a_t^2 - 4 * \lambda_h^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + 4 * \frac{\lambda_h * a_t^3 * \hat{\mu}}{\varepsilon} + 5 * \ln \frac{M_s^2}{\mu_M^2} \\
& * a_t^5 * \hat{\mu} * \hat{\mu}_h^2 - 5 * \kappa * a_t^3 * \hat{\mu} + 6 * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^3 * \hat{\mu} * \hat{\mu}_h^2 - 6 * \kappa^2 \\
& * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 + 6 * \frac{a_t^6 * \hat{\mu}_h^2}{\varepsilon} + 6 * \frac{\kappa^2 * a_t^2}{\varepsilon} + 6 * \frac{\kappa^2 * a_t^2 * \hat{\mu}_h^4}{\varepsilon} \\
& - 7 * a_t^5 * \hat{\mu} * \hat{\mu}_h^2 - 7 * \frac{a_t^5 * \hat{\mu}}{\varepsilon} - 8 * \lambda_h * \ln \frac{M_s^2}{\mu_M^2} * a_t^3 * \hat{\mu} * \hat{\mu}_h^2 + 8 * \lambda_h \\
& * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + 8 * \frac{\lambda_h^2 * a_t^2}{\varepsilon} - 9 * \ln \frac{M_s^2}{\mu_M^2} * a_t^6 - 10 * \lambda_h * \kappa * a_t^2 \\
& - 10 * \frac{a_t^5 * \hat{\mu} * \hat{\mu}_h^2}{\varepsilon} - 12 * \kappa * a_t^3 * \hat{\mu} * \hat{\mu}_h^2 + 12 * \lambda_h * a_t^3 * \hat{\mu} * \hat{\mu}_h^2 - 12 \\
& * \lambda_h * a_t^4 * \hat{\mu}_h^2 + 12 * \lambda_h * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 + 12 * \lambda_h * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 * \hat{\mu}_h^2 + 12 \\
& * \lambda_h^2 * a_t^2 * \hat{\mu}_h^2 - 12 * \lambda_h^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 - 12 * \frac{\kappa * a_t^3 * \hat{\mu} * \hat{\mu}_h^2}{\varepsilon} \\
& + 12 * \frac{\kappa^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} - 16 * \lambda_h * a_t^4 + 16 * \frac{\lambda_h * a_t^3 * \hat{\mu} * \hat{\mu}_h^2}{\varepsilon} - 16 \\
& * \frac{\lambda_h * \kappa * a_t^2}{\varepsilon} + 18 * \frac{a_t^6}{\varepsilon} + 20 * \lambda_h * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 + 21 * \frac{\kappa * a_t^4}{\varepsilon} \\
& - 24 * \lambda_h * \kappa * a_t^2 * \hat{\mu}_h^2 - 24 * \frac{\lambda_h * a_t^4}{\varepsilon} - 24 * \frac{\lambda_h * a_t^4 * \hat{\mu}_h^2}{\varepsilon} + 24 \\
& * \frac{\lambda_h^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} - 27 * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 * \hat{\mu}_h^2 + 37 * \kappa * a_t^4 * \hat{\mu}_h^2 - 40 \\
& * \frac{\lambda_h * \kappa * a_t^2 * \hat{\mu}_h^2}{\varepsilon} + 54 * \frac{\kappa * a_t^4 * \hat{\mu}_h^2}{1\mathbb{F}} - \frac{\lambda_\varphi * a_t^4 * \hat{\mu}_h^2}{\varepsilon} \Big) * M_s^{-2} \\
& \varepsilon_{i,l} * \varepsilon_{j,m} * \varepsilon_{k,n} * H_i * H_j * H_k * (H^\dagger)_l * (H^\dagger)_m * (H^\dagger)_n
\end{aligned} \tag{48}$$

term30

$$\begin{aligned}
& - \left(\lambda_h * \ln \frac{M_s^2}{\mu_M^2} * a_t^3 * \hat{\mu} - \lambda_h * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - 2 * \lambda_h * a_t^3 * \hat{\mu} + 2 * \lambda_h * \kappa * a_t^2 - 2 \right. \\
& * \lambda_h^2 * a_t^2 + 2 * \lambda_h^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - 2 * \frac{\lambda_h * a_t^3 * \hat{\mu}}{\varepsilon} + 2 * \frac{\lambda_h * \kappa * a_t^2}{\varepsilon} + 4 * \lambda_h * \ln \frac{M_s^2}{\mu_M^2} \\
& * a_t^3 * \hat{\mu} * \hat{\mu}_h^2 - 4 * \lambda_h * \kappa * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 - 4 * \frac{\lambda_h^2 * a_t^2}{\varepsilon} - 6 * \lambda_h * a_t^3 * \hat{\mu} * \hat{\mu}_h^2 + 6 \\
& * \lambda_h * a_t^4 * \hat{\mu}_h^2 - 6 * \lambda_h * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 - 6 * \lambda_h * \ln \frac{M_s^2}{\mu_M^2} * a_t^4 * \hat{\mu}_h^2 + 6 * \lambda_h * \kappa * a_t^2 * \hat{\mu}_h^2 \\
& - 6 * \lambda_h^2 * a_t^2 * \hat{\mu}_h^2 + 6 * \lambda_h^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 + 8 * \lambda_h * a_t^4 - 8 * \frac{\lambda_h * a_t^3 * \hat{\mu} * \hat{\mu}_h^2}{\varepsilon} + 8 \\
& * \frac{\lambda_h * \kappa * a_t^2 * \hat{\mu}_h^2}{\varepsilon} + 12 * \frac{\lambda_h * a_t^4}{\varepsilon} + 12 * \frac{\lambda_h * a_t^4 * \hat{\mu}_h^2}{\varepsilon} - 12 * \frac{\lambda_h^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} \Big) * M_s^{-2} \\
& \varepsilon_{i,m} * \varepsilon_{j,l} * \varepsilon_{k,n} * H_i * H_j * H_k * (H^\dagger)_l * (H^\dagger)_m * (H^\dagger)_n
\end{aligned} \tag{49}$$

term31

$$\begin{aligned}
& + \left(2 * \lambda_h^2 * a_t^2 - 2 * \lambda_h^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + 4 * \frac{\lambda_h^2 * a_t^2}{\varepsilon} + 6 * \lambda_h^2 * a_t^2 \right. \\
& * \hat{\mu}_h^2 - 6 * \lambda_h^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 + 12 * \frac{\lambda_h^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} \Big) * M_s^{-2} \\
& \varepsilon_{i,l} * \varepsilon_{j,n} * \varepsilon_{k,m} * H_i * H_j * H_k * (H^\dagger)_l * (H^\dagger)_m * (H^\dagger)_n
\end{aligned} \tag{50}$$

term32

$$\begin{aligned}
& + \left(\frac{1}{2} * \lambda_h^2 * a_t^2 - \frac{1}{2} * \lambda_h^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{3}{2} * \lambda_h^2 * a_t^2 * \hat{\mu}_h^2 - \frac{3}{2} \right. \\
& * \lambda_h^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 + 3 * \frac{\lambda_h^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} + \frac{\lambda_h^2 * a_t^2}{\varepsilon} \Big) * M_s^{-2} \\
& \varepsilon_{i,n} * \varepsilon_{j,l} * \varepsilon_{k,m} * H_i * H_j * H_k * (H^\dagger)_l * (H^\dagger)_m * (H^\dagger)_n
\end{aligned} \tag{51}$$

term33

$$\begin{aligned}
& + \left(\frac{1}{2} * \lambda_h^2 * a_t^2 - \frac{1}{2} * \lambda_h^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{3}{2} * \lambda_h^2 * a_t^2 * \hat{\mu}_h^2 - \frac{3}{2} \right. \\
& * \lambda_h^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 * \hat{\mu}_h^2 + 3 * \frac{\lambda_h^2 * a_t^2 * \hat{\mu}_h^2}{\varepsilon} + \frac{\lambda_h^2 * a_t^2}{\varepsilon} \Big) * M_s^{-2} \\
& \varepsilon_{i,m} * \varepsilon_{j,n} * \varepsilon_{k,l} * H_i * H_j * H_k * (H^\dagger)_l * (H^\dagger)_m * (H^\dagger)_n
\end{aligned} \tag{52}$$

nD: 0, H: 1, H+: 1, WL+: 2

term34

$$\begin{aligned}
& - \left(\frac{5}{1152} * g_2^2 * a_t^2 - \frac{1}{192} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{96} * \frac{g_2^2 * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& \varepsilon_{l,m} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_3} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_4} * \delta_{i,k} * \delta_{j,n} * H_i * (H^\dagger)_j * (W_R)_{k,l}^{\dot{\alpha}_1, \dot{\alpha}_2} * (W_R)_{m,n}^{\dot{\alpha}_3, \dot{\alpha}_4}
\end{aligned} \tag{53}$$

term34

$$\begin{aligned}
& - \left(\frac{5}{1152} * g_2^2 * a_t^2 - \frac{1}{192} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{96} * \frac{g_2^2 * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& \varepsilon_{l,m} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_4} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_3} * \delta_{i,k} * \delta_{j,n} * H_i * (H^\dagger)_j * (W_R)_{k,l}^{\dot{\alpha}_1, \dot{\alpha}_2} * (W_R)_{m,n}^{\dot{\alpha}_3, \dot{\alpha}_4}
\end{aligned} \tag{54}$$

term35

$$+ \left(\frac{5}{1152} * g_2^2 * a_t^2 - \frac{1}{192} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{96} * \frac{g_2^2 * a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (55)$$

$$\varepsilon_{k,n} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_3} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_4} * \delta_{i,m} * \delta_{j,l} * H_i * (H^\dagger)_j * (W_R)_{k,l}^{\dot{\alpha}_1, \dot{\alpha}_2} * (W_R)_{m,n}^{\dot{\alpha}_3, \dot{\alpha}_4}$$

term35

$$+ \left(\frac{5}{1152} * g_2^2 * a_t^2 - \frac{1}{192} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{96} * \frac{g_2^2 * a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (56)$$

$$\varepsilon_{k,n} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_4} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_3} * \delta_{i,m} * \delta_{j,l} * H_i * (H^\dagger)_j * (W_R)_{k,l}^{\dot{\alpha}_1, \dot{\alpha}_2} * (W_R)_{m,n}^{\dot{\alpha}_3, \dot{\alpha}_4}$$

nD: 0, WL: 2, H: 1, H+: 1

term34

$$- \left(\frac{5}{1152} * g_2^2 * a_t^2 - \frac{1}{192} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{96} * \frac{g_2^2 * a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (57)$$

$$\varepsilon_{l,m} * \varepsilon^{\alpha_1, \alpha_3} * \varepsilon^{\alpha_2, \alpha_4} * \delta_{i,k} * \delta_{j,n} * (W_L)_{k,l, \alpha_1, \alpha_2} * (W_L)_{m,n, \alpha_3, \alpha_4} * H_i * (H^\dagger)_j$$

term34

$$- \left(\frac{5}{1152} * g_2^2 * a_t^2 - \frac{1}{192} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{96} * \frac{g_2^2 * a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (58)$$

$$\varepsilon_{l,m} * \varepsilon^{\alpha_1, \alpha_4} * \varepsilon^{\alpha_2, \alpha_3} * \delta_{i,k} * \delta_{j,n} * (W_L)_{k,l, \alpha_1, \alpha_2} * (W_L)_{m,n, \alpha_3, \alpha_4} * H_i * (H^\dagger)_j$$

term35

$$+ \left(\frac{5}{1152} * g_2^2 * a_t^2 - \frac{1}{192} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{96} * \frac{g_2^2 * a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (59)$$

$$\varepsilon_{k,n} * \varepsilon^{\alpha_1, \alpha_3} * \varepsilon^{\alpha_2, \alpha_4} * \delta_{i,m} * \delta_{j,l} * (W_L)_{k,l, \alpha_1, \alpha_2} * (W_L)_{m,n, \alpha_3, \alpha_4} * H_i * (H^\dagger)_j$$

term35

$$+ \left(\frac{5}{1152} * g_2^2 * a_t^2 - \frac{1}{192} * g_2^2 * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 + \frac{1}{96} * \frac{g_2^2 * a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (60)$$

$$\varepsilon_{k,n} * \varepsilon^{\alpha_1, \alpha_4} * \varepsilon^{\alpha_2, \alpha_3} * \delta_{i,m} * \delta_{j,l} * (W_L)_{k,l, \alpha_1, \alpha_2} * (W_L)_{m,n, \alpha_3, \alpha_4} * H_i * (H^\dagger)_j$$

nD: 1, L: 1, H: 1, H+: 1, L+: 1

term38

$$- \left(\frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{2} * \frac{a_t^2}{\varepsilon} - \frac{5}{8} * a_t^2 \right) * M_s^{-2} \quad (61)$$

$$(y_e)^{w,u} * (y_e^\dagger)^{u,v} * \varepsilon_{i,k} * \varepsilon_{j,l} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * L_{\alpha_1, i}^v * (D_{\alpha_2}^{\dot{\alpha}_1} H)_j * (H^\dagger)_k * (L^\dagger)_l^{\dot{\alpha}_2, w}$$

term43

$$+ \left(\frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{2} * \frac{a_t^2}{\varepsilon} - \frac{5}{8} * a_t^2 \right) * M_s^{-2} \quad (62)$$

$$(y_e)^{w,u} * (y_e^\dagger)^{u,v} * \varepsilon_{i,k} * \varepsilon_{j,l} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * L_{\alpha_1, i}^v * H_j * (D_{\alpha_2}^{\dot{\alpha}_1} H^\dagger)_k * (L^\dagger)_l^{\dot{\alpha}_2, w}$$

term48

$$+ \left(\frac{1}{4} * a_t^2 \right) * M_s^{-2} \\ (y_e)^{w,u} * (y_e^\dagger)^{u,v} * \varepsilon_{i,k} * \varepsilon_{j,l} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * (D_{\alpha_1}^{\dot{\alpha}_1} L)_{\alpha_2, i}^v * H_j * (H^\dagger)_k * (L^\dagger)_l^{\dot{\alpha}_2, w} \quad (63)$$

term53

$$- \left(\frac{1}{4} * a_t^2 \right) * M_s^{-2} \\ (y_e)^{w,u} * (y_e^\dagger)^{u,v} * \varepsilon_{i,k} * \varepsilon_{j,l} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * L_{\alpha_1, i}^v * H_j * (H^\dagger)_k * (D_{\alpha_2}^{\dot{\alpha}_1} L^\dagger)_l^{\dot{\alpha}_2, w} \quad (64)$$

term58

$$- \left(\frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{8} * a_t^2 - \frac{1}{2} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\ (y_e)^{w,u} * (y_e^\dagger)^{u,v} * \varepsilon_{i,k} * \varepsilon_{j,l} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * L_{\alpha_1, i}^v * H_j * (H^\dagger)_k * (D_{\alpha_2}^{\dot{\alpha}_1} L^\dagger)_l^{\dot{\alpha}_2, w} \quad (65)$$

term63

$$+ \left(\frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{8} * a_t^2 - \frac{1}{2} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\ (y_e)^{w,u} * (y_e^\dagger)^{u,v} * \varepsilon_{i,k} * \varepsilon_{j,l} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * (D_{\alpha_1}^{\dot{\alpha}_1} L)_{\alpha_2, i}^v * H_j * (H^\dagger)_k * (L^\dagger)_l^{\dot{\alpha}_2, w} \quad (66)$$

nD: 1, Q: 1, H: 1, H+: 1, Q+: 1

term39

$$- \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} - \frac{5}{16} * a_t^2 \right) * M_s^{-2} \\ (y_d)^{w,u} * (y_d^\dagger)^{u,v} * \varepsilon_{i,k} * \varepsilon_{j,l} * \varepsilon_{a,b,c} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * Q_{\alpha_1, i, a}^v * (D_{\alpha_2}^{\dot{\alpha}_1} H)_j * (H^\dagger)_k * (Q^\dagger)_{l, b, c}^{\dot{\alpha}_2, w} \quad (67)$$

term44

$$+ \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} - \frac{5}{16} * a_t^2 \right) * M_s^{-2} \\ (y_d)^{w,u} * (y_d^\dagger)^{u,v} * \varepsilon_{i,k} * \varepsilon_{j,l} * \varepsilon_{a,b,c} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * Q_{\alpha_1, i, a}^v * H_j * (D_{\alpha_2}^{\dot{\alpha}_1} H^\dagger)_k * (Q^\dagger)_{l, b, c}^{\dot{\alpha}_2, w} \quad (68)$$

term49

$$- \left(\frac{1}{8} * a_t^2 \right) * M_s^{-2} \\ (y_d)^{w,u} * (y_d^\dagger)^{u,v} * \varepsilon_{i,k} * \varepsilon_{j,l} * \varepsilon_{a,b,c} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (D_{\alpha_1}^{\dot{\alpha}_1} Q)_{\alpha_2, i, a}^v * H_j * (H^\dagger)_k * (Q^\dagger)_{l, b, c}^{\dot{\alpha}_2, w} \quad (69)$$

term54

$$- \left(\frac{1}{8} * a_t^2 \right) * M_s^{-2} \\ (y_d)^{w,u} * (y_d^\dagger)^{u,v} * \varepsilon_{i,k} * \varepsilon_{j,l} * \varepsilon_{a,b,c} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * Q_{\alpha_1, i, a}^v * H_j * (H^\dagger)_k * (D_{\alpha_2}^{\dot{\alpha}_1} Q^\dagger)_{l, b, c}^{\dot{\alpha}_2, w} \quad (70)$$

term59

$$\begin{aligned}
& - \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& (y_d)^{w,u} * (y_d^\dagger)^{u,v} * \varepsilon_{i,k} * \varepsilon_{j,l} * \varepsilon_{a,b,c} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * Q_{\alpha_1, i, a}^v * H_j * (H^\dagger)_k * (D_{\alpha_2}^{\dot{\alpha}_1} Q^\dagger)_{l, b, c}^{\dot{\alpha}_2, w}
\end{aligned} \tag{71}$$

term64

$$\begin{aligned}
& - \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& (y_d)^{w,u} * (y_d^\dagger)^{u,v} * \varepsilon_{i,k} * \varepsilon_{j,l} * \varepsilon_{a,b,c} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (D_{\alpha_1}^{\dot{\alpha}_1} Q)_{\alpha_2, i, a}^v * H_j * (H^\dagger)_k * (Q^\dagger)_{l, b, c}^{\dot{\alpha}_2, w}
\end{aligned} \tag{72}$$

term88

$$\begin{aligned}
& - \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} - \frac{5}{16} * a_t^2 \right) * M_s^{-2} \\
& (y_u)^{w,u} * (y_u^\dagger)^{u,v} * \varepsilon_{i,j} * \varepsilon_{l,k} * \varepsilon_{a,b,c} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * Q_{\alpha_1, i, a}^v * (D_{\alpha_2}^{\dot{\alpha}_1} H)_j * (H^\dagger)_k * (Q^\dagger)_{l, b, c}^{\dot{\alpha}_2, w}
\end{aligned} \tag{73}$$

term89

$$\begin{aligned}
& - \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} - \frac{5}{16} * a_t^2 \right) * M_s^{-2} \\
& (y_u)^{w,u} * (y_u^\dagger)^{u,v} * \varepsilon_{i,j} * \varepsilon_{k,l} * \varepsilon_{a,b,c} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * Q_{\alpha_1, i, a}^v * H_j * (D_{\alpha_2}^{\dot{\alpha}_1} H^\dagger)_k * (Q^\dagger)_{l, b, c}^{\dot{\alpha}_2, w}
\end{aligned} \tag{74}$$

term90

$$\begin{aligned}
& + \left(\frac{1}{8} * a_t^2 \right) * M_s^{-2} \\
& (y_u)^{w,u} * (y_u^\dagger)^{u,v} * \varepsilon_{j,i} * \varepsilon_{k,l} * \varepsilon_{a,b,c} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (D_{\alpha_1}^{\dot{\alpha}_1} Q)_{\alpha_2, i, a}^v * H_j * (H^\dagger)_k * (Q^\dagger)_{l, b, c}^{\dot{\alpha}_2, w}
\end{aligned} \tag{75}$$

term91

$$\begin{aligned}
& + \left(\frac{1}{8} * a_t^2 \right) * M_s^{-2} \\
& (y_u)^{w,u} * (y_u^\dagger)^{u,v} * \varepsilon_{i,j} * \varepsilon_{l,k} * \varepsilon_{a,b,c} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * Q_{\alpha_1, i, a}^v * H_j * (H^\dagger)_k * (D_{\alpha_2}^{\dot{\alpha}_1} Q^\dagger)_{l, b, c}^{\dot{\alpha}_2, w}
\end{aligned} \tag{76}$$

term92

$$\begin{aligned}
& + \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& (y_u)^{w,u} * (y_u^\dagger)^{u,v} * \varepsilon_{j,i} * \varepsilon_{k,l} * \varepsilon_{a,b,c} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * Q_{\alpha_1, i, a}^v * H_j * (H^\dagger)_k * (D_{\alpha_2}^{\dot{\alpha}_1} Q^\dagger)_{l, b, c}^{\dot{\alpha}_2, w}
\end{aligned} \tag{77}$$

term93

$$\begin{aligned}
& - \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& (y_u)^{w,u} * (y_u^\dagger)^{u,v} * \varepsilon_{j,i} * \varepsilon_{k,l} * \varepsilon_{a,b,c} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * (D_{\alpha_1}^{\dot{\alpha}_1} Q)_{\alpha_2, i, a}^v * H_j * (H^\dagger)_k * (Q^\dagger)_{l, b, c}^{\dot{\alpha}_2, w}
\end{aligned} \tag{78}$$

nD: 1, eC: 1, H: 1, H+: 1, eC+: 1

term40

$$\begin{aligned}
& - \left(\frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{2} * \frac{a_t^2}{\varepsilon} - \frac{5}{8} * a_t^2 \right) * M_s^{-2} \\
& (y_e)^{u,w} * (y_e^\dagger)^{v,u} * \varepsilon_{i,j} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (e_{\mathbb{C}})_{\alpha_1}^v * (D_{\alpha_2}^{\dot{\alpha}_1} H)_i * (H^\dagger)_j * (e_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}
\end{aligned} \tag{79}$$

term46

$$\begin{aligned}
& + \left(\frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{2} * \frac{a_t^2}{\varepsilon} - \frac{5}{8} * a_t^2 \right) * M_s^{-2} \\
& (y_e)^{u,w} * (y_e^\dagger)^{v,u} * \varepsilon_{i,j} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (e_{\mathbb{C}})_{\alpha_1}^v * H_i * (D_{\alpha_2}^{\dot{\alpha}_1} H^\dagger)_j * (e_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}
\end{aligned} \tag{80}$$

term51

$$\begin{aligned}
& - \left(\frac{1}{4} * a_t^2 \right) * M_s^{-2} \\
& (y_e)^{u,w} * (y_e^\dagger)^{v,u} * \varepsilon_{i,j} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (e_{\mathbb{C}})_{\alpha_1}^v * H_i * (H^\dagger)_j * (D_{\alpha_2}^{\dot{\alpha}_1} e_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}
\end{aligned} \tag{81}$$

term55

$$\begin{aligned}
& + \left(\frac{1}{4} * a_t^2 \right) * M_s^{-2} \\
& (y_e)^{u,w} * (y_e^\dagger)^{v,u} * \varepsilon_{i,j} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (D_{\alpha_1}^{\dot{\alpha}_1} e_{\mathbb{C}})_{\alpha_2}^v * H_i * (H^\dagger)_j * (e_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}
\end{aligned} \tag{82}$$

term61

$$\begin{aligned}
& + \left(\frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{8} * a_t^2 - \frac{1}{2} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& (y_e)^{u,w} * (y_e^\dagger)^{v,u} * \varepsilon_{i,j} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * (D_{\alpha_1}^{\dot{\alpha}_1} e_{\mathbb{C}})_{\alpha_2}^v * H_i * (H^\dagger)_j * (e_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}
\end{aligned} \tag{83}$$

term65

$$\begin{aligned}
& - \left(\frac{1}{4} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{8} * a_t^2 - \frac{1}{2} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& (y_e)^{u,w} * (y_e^\dagger)^{v,u} * \varepsilon_{i,j} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (e_{\mathbb{C}})_{\alpha_1}^v * H_i * (H^\dagger)_j * (D_{\alpha_2}^{\dot{\alpha}_1} e_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}
\end{aligned} \tag{84}$$

nD: 1, dC: 1, H: 1, H+: 1, dC+: 1

term41

$$\begin{aligned}
& - \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} - \frac{5}{16} * a_t^2 \right) * M_s^{-2} \\
& (y_d)^{u,w} * (y_d^\dagger)^{v,u} * \varepsilon_{i,j} * \varepsilon_{c,a,b} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (d_{\mathbb{C}})_{\alpha_1, a, b}^v * (D_{\alpha_2}^{\dot{\alpha}_1} H)_i * (H^\dagger)_j * (d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, w}
\end{aligned} \tag{85}$$

term47

$$+ \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} - \frac{5}{16} * a_t^2 \right) * M_s^{-2} \quad (86)$$

$$(y_d)^{u,w} * (y_d^\dagger)^{v,u} * \varepsilon_{i,j} * \varepsilon_{c,a,b} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (d_{\mathbb{C}})_{\alpha_1, a, b}^v * H_i * (D_{\alpha_2}^{\dot{\alpha}_1} H^\dagger)_j * (d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, w}$$

term52

$$- \left(\frac{1}{8} * a_t^2 \right) * M_s^{-2} \quad (87)$$

$$(y_d)^{u,w} * (y_d^\dagger)^{v,u} * \varepsilon_{i,j} * \varepsilon_{c,a,b} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (d_{\mathbb{C}})_{\alpha_1, a, b}^v * H_i * (H^\dagger)_j * (D_{\alpha_2}^{\dot{\alpha}_1} d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, w}$$

term56

$$+ \left(\frac{1}{8} * a_t^2 \right) * M_s^{-2} \quad (88)$$

$$(y_d)^{u,w} * (y_d^\dagger)^{v,u} * \varepsilon_{i,j} * \varepsilon_{c,a,b} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * (D_{\alpha_1}^{\dot{\alpha}_1} d_{\mathbb{C}})_{\alpha_2, a, b}^v * H_i * (H^\dagger)_j * (d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, w}$$

term62

$$+ \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (89)$$

$$(y_d)^{u,w} * (y_d^\dagger)^{v,u} * \varepsilon_{i,j} * \varepsilon_{c,a,b} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (D_{\alpha_1}^{\dot{\alpha}_1} d_{\mathbb{C}})_{\alpha_2, a, b}^v * H_i * (H^\dagger)_j * (d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, w}$$

term66

$$- \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (90)$$

$$(y_d)^{u,w} * (y_d^\dagger)^{v,u} * \varepsilon_{i,j} * \varepsilon_{c,a,b} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (d_{\mathbb{C}})_{\alpha_1, a, b}^v * H_i * (H^\dagger)_j * (D_{\alpha_2}^{\dot{\alpha}_1} d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, w}$$

nD: 1, uC: 1, H: 1, H+: 1, uC+: 1

term42

$$+ \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} - \frac{5}{16} * a_t^2 \right) * M_s^{-2} \quad (91)$$

$$(y_u)^{u,w} * (y_u^\dagger)^{v,u} * \varepsilon_{i,j} * \varepsilon_{c,a,b} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (u_{\mathbb{C}})_{\alpha_1, a, b}^v * (D_{\alpha_2}^{\dot{\alpha}_1} H)_i * (H^\dagger)_j * (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, w}$$

term45

$$- \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} - \frac{5}{16} * a_t^2 \right) * M_s^{-2} \quad (92)$$

$$(y_u)^{u,w} * (y_u^\dagger)^{v,u} * \varepsilon_{i,j} * \varepsilon_{c,a,b} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (u_{\mathbb{C}})_{\alpha_1, a, b}^v * H_i * (D_{\alpha_2}^{\dot{\alpha}_1} H^\dagger)_j * (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, w}$$

term50

$$- \left(\frac{1}{8} * a_t^2 \right) * M_s^{-2} \quad (93)$$

$$(y_u)^{u,w} * (y_u^\dagger)^{v,u} * \varepsilon_{i,j} * \varepsilon_{c,a,b} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (u_{\mathbb{C}})_{\alpha_1, a, b}^v * H_i * (H^\dagger)_j * (D_{\alpha_2}^{\dot{\alpha}_1} u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, w}$$

term57

$$+ \left(\frac{1}{8} * a_t^2 \right) * M_s^{-2} \quad (94)$$

$$(y_u)^{u,w} * (y_u^\dagger)^{v,u} * \varepsilon_{i,j} * \varepsilon_{c,a,b} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * (D_{\alpha_1}^{\dot{\alpha}_1} u_{\mathbb{C}})^v_{\alpha_2, a, b} * H_i * (H^\dagger)_j * (u_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c$$

term60

$$+ \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (95)$$

$$(y_u)^{u,w} * (y_u^\dagger)^{v,u} * \varepsilon_{i,j} * \varepsilon_{c,a,b} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (D_{\alpha_1}^{\dot{\alpha}_1} u_{\mathbb{C}})^v_{\alpha_2, a, b} * H_i * (H^\dagger)_j * (u_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c$$

term67

$$- \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (96)$$

$$(y_u)^{u,w} * (y_u^\dagger)^{v,u} * \varepsilon_{i,j} * \varepsilon_{c,a,b} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (u_{\mathbb{C}})^v_{\alpha_1, a, b} * H_i * (H^\dagger)_j * (D_{\alpha_2}^{\dot{\alpha}_1} u_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c$$

nD: 1, uC: 1, H: 2, dC+: 1

term68

$$- \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} - \frac{5}{16} * a_t^2 \right) * M_s^{-2} \quad (97)$$

$$(y_u^\dagger)^{v,u} * (y_d)^{u,w} * \varepsilon_{i,j} * \varepsilon_{c,a,b} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (u_{\mathbb{C}})^v_{\alpha_1, a, b} * (D_{\alpha_2}^{\dot{\alpha}_1} H)_i * H_j * (d_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c$$

term69

$$- \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} - \frac{5}{16} * a_t^2 \right) * M_s^{-2} \quad (98)$$

$$(y_u^\dagger)^{v,u} * (y_d)^{u,w} * \varepsilon_{i,j} * \varepsilon_{c,a,b} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (u_{\mathbb{C}})^v_{\alpha_1, a, b} * (D_{\alpha_2}^{\dot{\alpha}_1} H)_i * H_j * (d_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c$$

term73

$$- \left(\frac{1}{8} * a_t^2 \right) * M_s^{-2} \quad (99)$$

$$(y_u^\dagger)^{v,u} * (y_d)^{u,w} * \varepsilon_{i,j} * \varepsilon_{c,a,b} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (u_{\mathbb{C}})^v_{\alpha_1, a, b} * H_i * H_j * (D_{\alpha_2}^{\dot{\alpha}_1} d_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c$$

term74

$$- \left(\frac{1}{8} * a_t^2 \right) * M_s^{-2} \quad (100)$$

$$(y_u^\dagger)^{v,u} * (y_d)^{u,w} * \varepsilon_{i,j} * \varepsilon_{c,a,b} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * (D_{\alpha_1}^{\dot{\alpha}_1} u_{\mathbb{C}})^v_{\alpha_2, a, b} * H_i * H_j * (d_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c$$

term77

$$- \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (101)$$

$$(y_u^\dagger)^{v,u} * (y_d)^{u,w} * \varepsilon_{j,i} * \varepsilon_{c,a,b} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (D_{\alpha_1}^{\dot{\alpha}_1} u_{\mathbb{C}})^v_{\alpha_2, a, b} * H_i * H_j * (d_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, w}_c$$

term78

$$\begin{aligned}
& - \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& (y_u^\dagger)^{v,u} * (y_d)^{u,w} * \varepsilon_{j,i} * \varepsilon_{c,a,b} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (u_{\mathbb{C}})_{\alpha_1, a, b}^v * H_i * H_j * (D_{\alpha_2}^{\dot{\alpha}_1} d_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, w}
\end{aligned} \tag{102}$$

nD: 1, dC: 1, H+: 2, uC+: 1

term70

$$\begin{aligned}
& - \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} - \frac{5}{16} * a_t^2 \right) * M_s^{-2} \\
& (y_u)^{u,w} * (y_d^\dagger)^{v,u} * \varepsilon_{i,j} * \varepsilon_{c,a,b} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (d_{\mathbb{C}})_{\alpha_1, a, b}^v * (D_{\alpha_2}^{\dot{\alpha}_1} H^\dagger)_i * (H^\dagger)_j * (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, w}
\end{aligned} \tag{103}$$

term71

$$\begin{aligned}
& + \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} - \frac{5}{16} * a_t^2 \right) * M_s^{-2} \\
& (y_u)^{u,w} * (y_d^\dagger)^{v,u} * \varepsilon_{j,i} * \varepsilon_{c,a,b} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (d_{\mathbb{C}})_{\alpha_1, a, b}^v * (D_{\alpha_2}^{\dot{\alpha}_1} H^\dagger)_i * (H^\dagger)_j * (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, w}
\end{aligned} \tag{104}$$

term72

$$\begin{aligned}
& - \left(\frac{1}{8} * a_t^2 \right) * M_s^{-2} \\
& (y_u)^{u,w} * (y_d^\dagger)^{v,u} * \varepsilon_{i,j} * \varepsilon_{c,a,b} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (d_{\mathbb{C}})_{\alpha_1, a, b}^v * (H^\dagger)_i * (H^\dagger)_j * (D_{\alpha_2}^{\dot{\alpha}_1} u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, w}
\end{aligned} \tag{105}$$

term75

$$\begin{aligned}
& - \left(\frac{1}{8} * a_t^2 \right) * M_s^{-2} \\
& (y_u)^{u,w} * (y_d^\dagger)^{v,u} * \varepsilon_{i,j} * \varepsilon_{c,a,b} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * (D_{\alpha_1}^{\dot{\alpha}_1} d_{\mathbb{C}})_{\alpha_2, a, b}^v * (H^\dagger)_i * (H^\dagger)_j * (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, w}
\end{aligned} \tag{106}$$

term76

$$\begin{aligned}
& + \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& (y_u)^{u,w} * (y_d^\dagger)^{v,u} * \varepsilon_{i,j} * \varepsilon_{c,a,b} * \varepsilon^{\alpha_2, \alpha_1} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (D_{\alpha_1}^{\dot{\alpha}_1} d_{\mathbb{C}})_{\alpha_2, a, b}^v * (H^\dagger)_i * (H^\dagger)_j * (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, w}
\end{aligned} \tag{107}$$

term79

$$\begin{aligned}
& + \left(\frac{1}{8} * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * a_t^2 - \frac{1}{4} * \frac{a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& (y_u)^{u,w} * (y_d^\dagger)^{v,u} * \varepsilon_{i,j} * \varepsilon_{c,a,b} * \varepsilon^{\alpha_1, \alpha_2} * \varepsilon_{\dot{\alpha}_1, \dot{\alpha}_2} * (d_{\mathbb{C}})_{\alpha_1, a, b}^v * (H^\dagger)_i * (H^\dagger)_j * (D_{\alpha_2}^{\dot{\alpha}_1} u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, w}
\end{aligned} \tag{108}$$

nD: 0, L: 1, eC: 1, H: 1, H+: 2

term80

$$+ \left(\frac{1}{8} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * d * a_t^2 - \frac{1}{4} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (109)$$

$$(y_e)^{u,v} * (y_e^\dagger)^{x,u} * (y_e^\dagger)^{v,w} * \varepsilon_{i,l} * \varepsilon_{j,k} * \varepsilon^{\alpha_1, \alpha_2} * L_{\alpha_1, i}^w * (e_{\mathbb{C}})_{\alpha_2}^x * H_j * (H^\dagger)_k * (H^\dagger)_l$$

term86

$$+ \left(\frac{1}{8} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * d * a_t^2 - \frac{1}{4} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (110)$$

$$(y_e)^{u,v} * (y_e^\dagger)^{x,u} * (y_e^\dagger)^{v,w} * \varepsilon_{i,k} * \varepsilon_{j,l} * \varepsilon^{\alpha_1, \alpha_2} * L_{\alpha_1, i}^w * (e_{\mathbb{C}})_{\alpha_2}^x * H_j * (H^\dagger)_k * (H^\dagger)_l$$

nD: 0, Q: 1, dC: 1, H: 1, H+: 2

term81

$$+ \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (111)$$

$$(y_d)^{u,v} * (y_d^\dagger)^{x,u} * (y_d^\dagger)^{v,w} * \varepsilon_{i,l} * \varepsilon_{j,k} * \varepsilon_{a,b,c} * \varepsilon^{\alpha_1, \alpha_2} * Q_{\alpha_1, i, a}^w * (d_{\mathbb{C}})_{\alpha_2, b, c}^x * H_j * (H^\dagger)_k * (H^\dagger)_l$$

term87

$$+ \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (112)$$

$$(y_d)^{u,v} * (y_d^\dagger)^{x,u} * (y_d^\dagger)^{v,w} * \varepsilon_{i,k} * \varepsilon_{j,l} * \varepsilon_{a,b,c} * \varepsilon^{\alpha_1, \alpha_2} * Q_{\alpha_1, i, a}^w * (d_{\mathbb{C}})_{\alpha_2, b, c}^x * H_j * (H^\dagger)_k * (H^\dagger)_l$$

term102

$$+ \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (113)$$

$$(y_u)^{v,u} * (y_u^\dagger)^{u,w} * (y_d^\dagger)^{x,v} * \varepsilon_{i,j} * \varepsilon_{k,l} * \varepsilon_{a,b,c} * \varepsilon^{\alpha_1, \alpha_2} * Q_{\alpha_1, i, a}^w * (d_{\mathbb{C}})_{\alpha_2, b, c}^x * H_j * (H^\dagger)_k * (H^\dagger)_l$$

term105

$$- \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (114)$$

$$(y_u)^{v,u} * (y_u^\dagger)^{u,w} * (y_d^\dagger)^{x,v} * \varepsilon_{i,j} * \varepsilon_{k,l} * \varepsilon_{a,b,c} * \varepsilon^{\alpha_1, \alpha_2} * Q_{\alpha_1, i, a}^w * (d_{\mathbb{C}})_{\alpha_2, b, c}^x * H_j * (H^\dagger)_k * (H^\dagger)_l$$

nD: 0, H: 2, H+: 1, L+: 1, eC+: 1

term82

$$- \left(\frac{1}{8} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * d * a_t^2 - \frac{1}{4} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \quad (115)$$

$$(y_e)^{w,u} * (y_e)^{v,x} * (y_e^\dagger)^{u,v} * \varepsilon_{i,l} * \varepsilon_{j,k} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * H_i * H_j * (H^\dagger)_k * (L^\dagger)_l^{\dot{\alpha}_1, w} * (e_{\mathbb{C}}^\dagger)^{\dot{\alpha}_2, x}$$

term84

$$\begin{aligned}
& - \left(\frac{1}{8} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{16} * d * a_t^2 - \frac{1}{4} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& (y_e)^{w,u} * (y_e)^{v,x} * (y_e^\dagger)^{u,v} * \varepsilon_{i,k} * \varepsilon_{j,l} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * H_i * H_j * (H^\dagger)_k * (L^\dagger)_l^{\dot{\alpha}_1, w} * (e_C^\dagger)^{\dot{\alpha}_2, x}
\end{aligned} \tag{116}$$

nD: 0, H: 2, H+: 1, Q+: 1, dC+: 1

term83

$$\begin{aligned}
& - \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& (y_d)^{w,u} * (y_d)^{v,x} * (y_d^\dagger)^{u,v} * \varepsilon_{i,l} * \varepsilon_{j,k} * \varepsilon_{c,a,b} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * H_i * H_j * (H^\dagger)_k * (Q^\dagger)_{l,a,b}^{\dot{\alpha}_1, w} * (d_C^\dagger)_c^{\dot{\alpha}_2, x}
\end{aligned} \tag{117}$$

term85

$$\begin{aligned}
& - \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& (y_d)^{w,u} * (y_d)^{v,x} * (y_d^\dagger)^{u,v} * \varepsilon_{i,k} * \varepsilon_{j,l} * \varepsilon_{c,a,b} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * H_i * H_j * (H^\dagger)_k * (Q^\dagger)_{l,a,b}^{\dot{\alpha}_1, w} * (d_C^\dagger)_c^{\dot{\alpha}_2, x}
\end{aligned} \tag{118}$$

term103

$$\begin{aligned}
& + \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& (y_u)^{w,u} * (y_u^\dagger)^{u,v} * (y_d)^{v,x} * \varepsilon_{j,i} * \varepsilon_{k,l} * \varepsilon_{c,a,b} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * H_i * H_j * (H^\dagger)_k * (Q^\dagger)_{l,a,b}^{\dot{\alpha}_1, w} * (d_C^\dagger)_c^{\dot{\alpha}_2, x}
\end{aligned} \tag{119}$$

term104

$$\begin{aligned}
& + \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& (y_u)^{w,u} * (y_u^\dagger)^{u,v} * (y_d)^{v,x} * \varepsilon_{i,j} * \varepsilon_{k,l} * \varepsilon_{c,a,b} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * H_i * H_j * (H^\dagger)_k * (Q^\dagger)_{l,a,b}^{\dot{\alpha}_1, w} * (d_C^\dagger)_c^{\dot{\alpha}_2, x}
\end{aligned} \tag{120}$$

nD: 0, H: 1, H+: 2, Q+: 1, uC+: 1

term94

$$\begin{aligned}
& - \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& (y_u)^{w,u} * (y_u)^{v,x} * (y_u^\dagger)^{u,v} * \varepsilon_{i,j} * \varepsilon_{k,l} * \varepsilon_{c,a,b} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * H_i * (H^\dagger)_j * (H^\dagger)_k * (Q^\dagger)_{l,a,b}^{\dot{\alpha}_1, w} * (u_C^\dagger)_c^{\dot{\alpha}_2, x}
\end{aligned} \tag{121}$$

term95

$$\begin{aligned}
& - \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& (y_u)^{v,x} * (y_d)^{w,u} * (y_d^\dagger)^{u,v} * \varepsilon_{i,l} * \varepsilon_{k,j} * \varepsilon_{c,a,b} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * H_i * (H^\dagger)_j * (H^\dagger)_k * (Q^\dagger)_{l,a,b}^{\dot{\alpha}_1, w} * (u_C^\dagger)_c^{\dot{\alpha}_2, x}
\end{aligned} \tag{122}$$

term99

$$\begin{aligned}
& - \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& (y_u)^{w,u} * (y_u)^{v,x} * (y_u^\dagger)^{u,v} * \varepsilon_{i,k} * \varepsilon_{j,l} * \varepsilon_{c,a,b} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * H_i * (H^\dagger)_j * (H^\dagger)_k * (Q^\dagger)_{l,a,b}^{\dot{\alpha}_1, w} * (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, x}
\end{aligned} \tag{123}$$

term101

$$\begin{aligned}
& - \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& (y_u)^{v,x} * (y_d)^{w,u} * (y_d^\dagger)^{u,v} * \varepsilon_{i,l} * \varepsilon_{j,k} * \varepsilon_{c,a,b} * \varepsilon_{\dot{\alpha}_2, \dot{\alpha}_1} * H_i * (H^\dagger)_j * (H^\dagger)_k * (Q^\dagger)_{l,a,b}^{\dot{\alpha}_1, w} * (u_{\mathbb{C}}^\dagger)_c^{\dot{\alpha}_2, x}
\end{aligned} \tag{124}$$

nD: 0, Q: 1, uC: 1, H: 2, H+: 1

term96

$$\begin{aligned}
& - \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& (y_u^\dagger)^{x,u} * (y_d)^{u,v} * (y_d^\dagger)^{v,w} * \varepsilon_{i,l} * \varepsilon_{j,k} * \varepsilon_{a,b,c} * \varepsilon^{\alpha_1, \alpha_2} * Q_{\alpha_1, i, a}^w * (u_{\mathbb{C}})_{\alpha_2, b, c}^x * H_j * H_k * (H^\dagger)_l
\end{aligned} \tag{125}$$

term97

$$\begin{aligned}
& + \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& (y_u)^{u,v} * (y_u^\dagger)^{x,u} * (y_u^\dagger)^{v,w} * \varepsilon_{i,k} * \varepsilon_{j,l} * \varepsilon_{a,b,c} * \varepsilon^{\alpha_1, \alpha_2} * Q_{\alpha_1, i, a}^w * (u_{\mathbb{C}})_{\alpha_2, b, c}^x * H_j * H_k * (H^\dagger)_l
\end{aligned} \tag{126}$$

term98

$$\begin{aligned}
& + \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& (y_u)^{u,v} * (y_u^\dagger)^{x,u} * (y_u^\dagger)^{v,w} * \varepsilon_{i,j} * \varepsilon_{k,l} * \varepsilon_{a,b,c} * \varepsilon^{\alpha_1, \alpha_2} * Q_{\alpha_1, i, a}^w * (u_{\mathbb{C}})_{\alpha_2, b, c}^x * H_j * H_k * (H^\dagger)_l
\end{aligned} \tag{127}$$

term100

$$\begin{aligned}
& - \left(\frac{1}{16} * d * \ln \frac{M_s^2}{\mu_M^2} * a_t^2 - \frac{3}{32} * d * a_t^2 - \frac{1}{8} * \frac{d * a_t^2}{\varepsilon} \right) * M_s^{-2} \\
& (y_u^\dagger)^{x,u} * (y_d)^{u,v} * (y_d^\dagger)^{v,w} * \varepsilon_{i,l} * \varepsilon_{k,j} * \varepsilon_{a,b,c} * \varepsilon^{\alpha_1, \alpha_2} * Q_{\alpha_1, i, a}^w * (u_{\mathbb{C}})_{\alpha_2, b, c}^x * H_j * H_k * (H^\dagger)_l
\end{aligned} \tag{128}$$